

Assignment 6: Data Scraping

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on data scraping.

Directions

1. Rename this file `<FirstLast>_A10_DataScraping.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Load the packages `tidyverse`, `rvest`, and any others you end up using.
 - Check your working directory

```
#1 loading tidyverse, rvest, and here
library(tidyverse)
library(rvest)
library(here)
library(ggthemes)
library(cowplot)

#verifying my working directory
here()
```

```
## [1] "/home/guest/EDE_R_Lab_F26_NEW"
```

```
#Establishing my color palette
canva_pal("Birds and berries")
```

```
## function (n)
## {
##   n_values <- length(values)
##   if (n > n_values) {
##     cli::cli_warn("This manual palette can handle a maximum of {n_values} values. You have suppl
```

```
##     }
##     unname(values[seq_len(n)])
## }
## <bytecode: 0x5ec8d11905a8>
## <environment: 0x5ec8d118fd90>
```

```
#Establishing my theme
my.theme <- theme_bw() +
  #making the title bold, 13 font, and centered
  theme(plot.title = element_text(face = "bold", size = 14, hjust = 0.5),
    #making my axis titles bold and 10 font
    axis.title = element_text(face = "bold", size = 10),
    #positioning my legend on the right side of the plot
    legend.position = "right"
  )

#setting my theme
theme_set(my.theme)
```

2. We will be scraping data from the NC DEQs Local Water Supply Planning website, specifically the Durham’s 2024 Municipal Local Water Supply Plan (LWSP):

- Navigate to <https://www.ncwater.org/WUDC/app/LWSP/search.php>
- Scroll down and select the LWSP link next to Durham Municipality.
- Note the web address: <https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024>

Indicate this website as the as the URL to be scraped. (In other words, read the contents into an `rvest` webpage object.)

```
#2
Durham_LWSP_Website <- read_html('https://www.ncwater.org/WUDC/app/LWSP/report.php?pwsid=03-32-010&year=2024')

#I know the full website isn't readable, but I couldn't get it to bump down to the
#next line and the code still work.
```

3. The data we want to collect are listed below:

- From the “1. System Information” section:
 - Water system name
 - PWSID
 - Ownership
- From the “3. Water Supply Sources” section:
 - Maximum Day Use (MGD) - for each month

In the code chunk below scrape these values, assigning them to four separate variables.

HINT: The first value should be “Durham”, the second “03-32-010”, the third “Municipality”, and the last should be a vector of 12 numeric values (represented as strings)“.

```

#3 creating the water system name variable
water_system_name <- Durham_LWSP_Website %>%
  html_nodes("div+ table tr:nth-child(1) td:nth-child(2)") %>%
  html_text()

#creating the PWSID variable
PWSID <- Durham_LWSP_Website %>%
  html_nodes("td tr:nth-child(1) td:nth-child(5)") %>%
  html_text()

#creating the ownership variable
ownership <- Durham_LWSP_Website %>%
  html_nodes("div+ table tr:nth-child(2) td:nth-child(4)") %>%
  html_text()

#creating the maximum daily use for each month of 2024 variable
maximum_daily_use_monthly <- Durham_LWSP_Website %>%
  html_nodes("th~ td+ td") %>%
  html_text()

```

4. Convert your scraped data into a dataframe. This dataframe should have a column for each of the 4 variables scraped and a row for the month corresponding to the withdrawal data. Also add a Date column that includes your month and year in data format. (Feel free to add a Year column too, if you wish.)

TIP: Use `rep()` to repeat a value when creating a dataframe.

NOTE: It's likely you won't be able to scrape the monthly withdrawal data in chronological order. You can overcome this by creating a month column manually assigning values in the order the data are scraped: "Jan", "May", "Sept", "Feb", etc...

5. Create a line plot of the maximum daily withdrawals across the months for 2024, making sure, the months are presented in proper sequence.

```

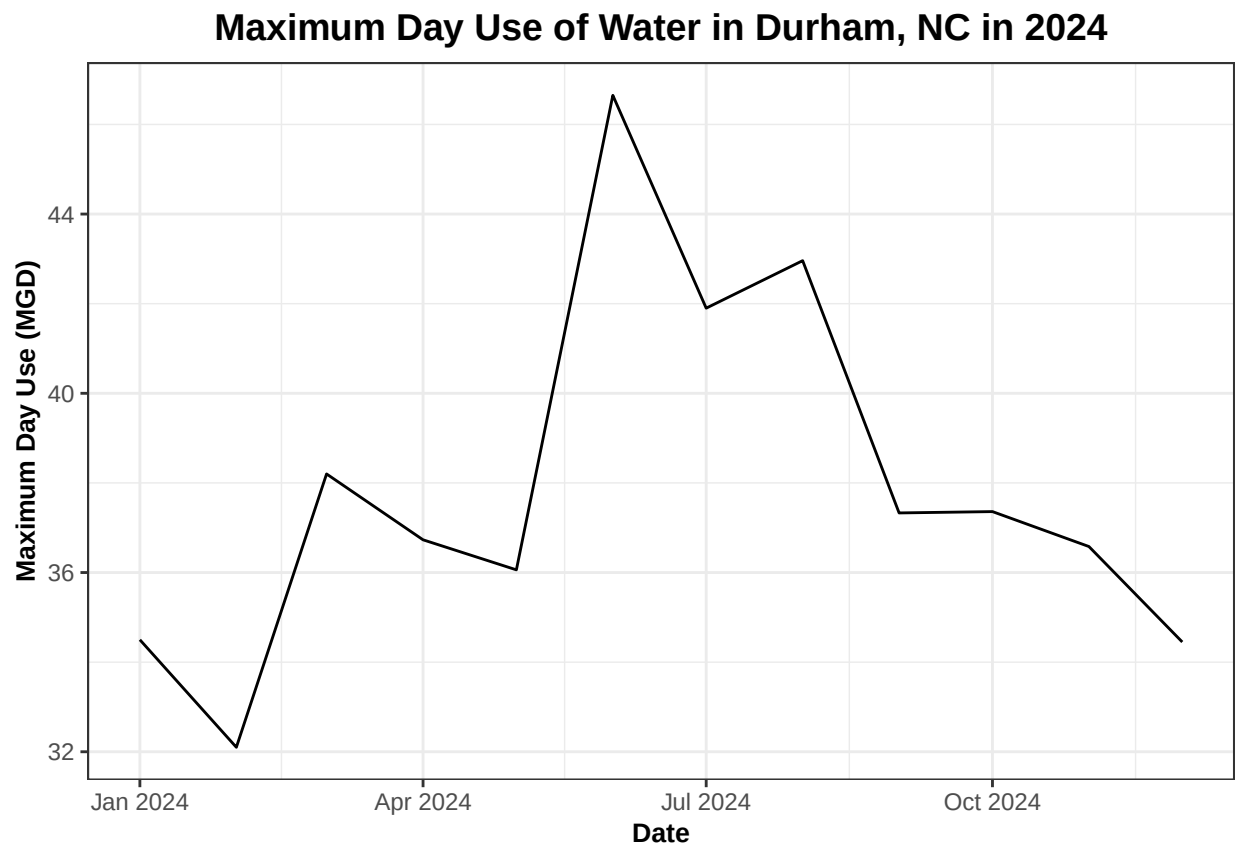
#4 creating a dataframe with the columns of the four variables I just scraped and a
#row for the months
Durham_2024_LWSP <- data.frame("water_system_name" = as.character(water_system_name),
                              "PWSID" = as.character(PWSID),
                              "ownership" = as.character(ownership),
                              "maximum_daily_use_monthly" =
                                as.numeric(maximum_daily_use_monthly))

#adding a date column to the dataframe
Durham_2024_LWSP <- Durham_2024_LWSP %>%
  mutate(Date = my(paste(c("Jan", "May", "Sept", "Feb", "June", "Oct", "Mar", "Jul",
                          "Nov", "Apr", "Aug", "Dec"), 2024))) %>%
  mutate(Year = year(Date))

#viewing the dataframe and commenting it out after
#view(Durham_2024_LWSP)

```

```
#5 creating a line plot for the maxium daily use values for each month in 2024
ggplot(Durham_2024_LWSP, aes(x=Date, y=maximum_daily_use_monthly)) +
  geom_line() +
  labs(title = "Maximum Day Use of Water in Durham, NC in 2024",
       x= "Date",
       y="Maximum Day Use (MGD)") +
  my.theme
```



6. Note that the PWSID and the year appear in the web address for the page we scraped. Construct a function with two inputs - “PWSID” and “year” - that:
 - Creates a URL pointing to the LWSP for that PWSID for the given year
 - Creates a website object and scrapes the data from that object (just as you did above)
 - Constructs a dataframe from the scraped data, mostly as you did above, but includes the PWSID and year provided as function inputs in the dataframe.
 - Returns the dataframe as the function’s output

```

#6 creating a scraping function with PWSID and year
scrape.it.LWSP <- function(PWSID, Year){

  #retrieving the LWSP website contents
  NC_LWSP_Website <- read_html(paste0(
'https://www.ncwater.org/WUDC/app/LWSP/report.php?', 'pwsid=', PWSID, '&year=', Year))

  #setting the element variables
  PWSID_tag <- 'td tr:nth-child(1) td:nth-child(5)'
  Year_tag <- '.fancy-table:nth-child(64) tr:nth-child(1) th:nth-child(2)'
  water_system_name_tag <- 'div+ table tr:nth-child(1) td:nth-child(2)'
  ownership_tag <- 'div+ table tr:nth-child(2) td:nth-child(4)'
  maximum_daily_use_monthly_tag <- 'th~ td+ td'

  #scraping the data items
  PWSID_website <- NC_LWSP_Website %>%
    html_nodes(PWSID_tag) %>%
    html_text()

  Year_website <- NC_LWSP_Website %>%
    html_nodes(Year_tag) %>%
    html_text()

  water_system_name_website <- NC_LWSP_Website %>%
    html_nodes(water_system_name_tag) %>%
    html_text()

  ownership_website <- NC_LWSP_Website %>%
    html_nodes(ownership_tag) %>%
    html_text()

  maximum_daily_use_website <- NC_LWSP_Website %>%
    html_nodes(maximum_daily_use_monthly_tag) %>%
    html_text()

  #creating the dataframe
  NC_LWSP <- data.frame("water_system_name" = as.character(water_system_name_website),
                        "ownership" = as.character(ownership_website),
                        "maximum_daily_use_monthly" =
                          as.numeric(maximum_daily_use_website)) %>%
    mutate(PWSID = !!PWSID,
           Year = !!Year,
           Date = my(paste(c("Jan", "May", "Sept", "Feb", "June", "Oct",
                             "Mar", "Jul", "Nov", "Apr", "Aug", "Dec"),
                           Year)))

  #returns the dataframe as the function's output
  return(NC_LWSP)
}

```

7. Use the function above to extract and plot max daily withdrawals for Durham (PWSID='03-32-010') for each month in 2020. Then show the structure of the dataframe with either the `str()` or the `glimpse()` function.

```

#7 using my scrape.it function to extract the max daily use for Durham for each
#month in 2020 and return it in a dataframe
Durham_2020_LWSP <- scrape.it.LWSP('03-32-010', 2020)

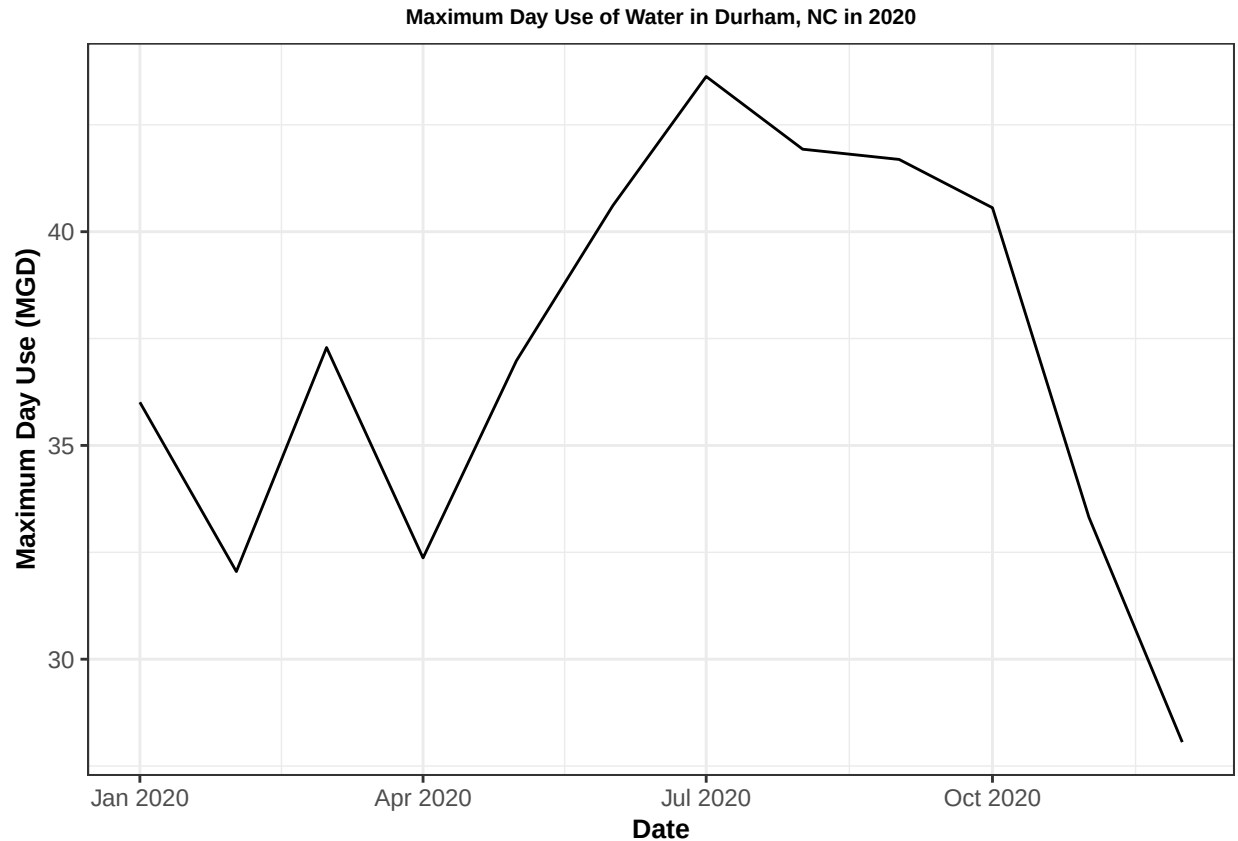
#viewing the structure of my dataframe
glimpse(Durham_2020_LWSP)

## Rows: 12
## Columns: 6
## $ water_system_name      <chr> "Durham", "Durham", "Durham", "Durham", "Dur~
## $ ownership              <chr> "Municipality", "Municipality", "Municipalit~
## $ maximum_daily_use_monthly <dbl> 36.01, 36.98, 41.69, 32.05, 40.61, 40.56, 37~
## $ PWSID                  <chr> "03-32-010", "03-32-010", "03-32-010", "03-3~
## $ Year                   <dbl> 2020, 2020, 2020, 2020, 2020, 2020, 2020, 20~
## $ Date                   <date> 2020-01-01, 2020-05-01, 2020-09-01, 2020-02~

#plotting the date and maximum day use of water from the dataframe
Durham_2020_Plot <- Durham_2020_LWSP %>%
ggplot(aes(x= Date, y= maximum_daily_use_monthly)) +
  geom_line() +
  labs(title = "Maximum Day Use of Water in Durham, NC in 2020",
       x= "Date",
       y="Maximum Day Use (MGD)") +
  my.theme +
  theme(plot.title = element_text(face = "bold", size = 8, hjust = 0.5))

#viewing the plot
Durham_2020_Plot

```

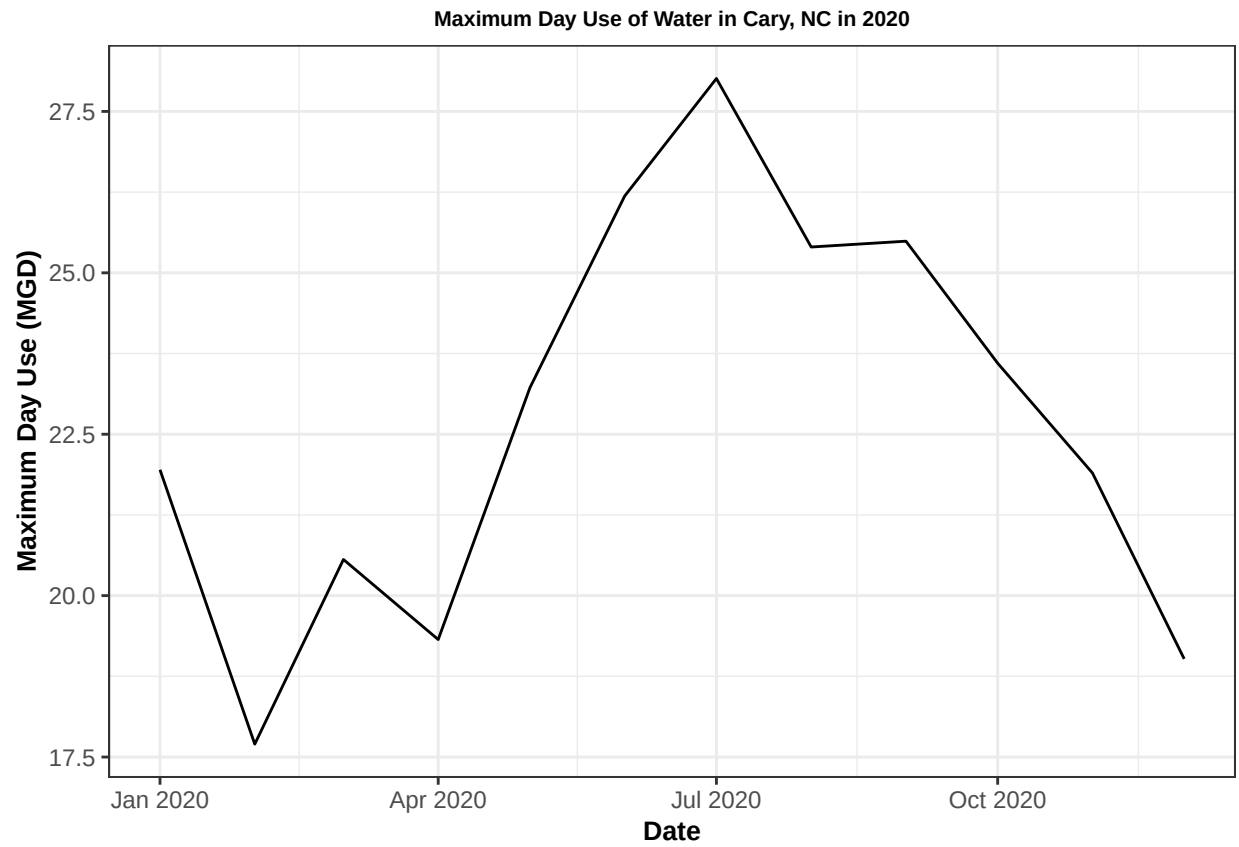


- Use the function above to extract data for Cary (PWSID = '03-32-010') in 2020. Combine this data with the Durham data collected above and create a plot that compares Cary's to Durham's water withdrawals.

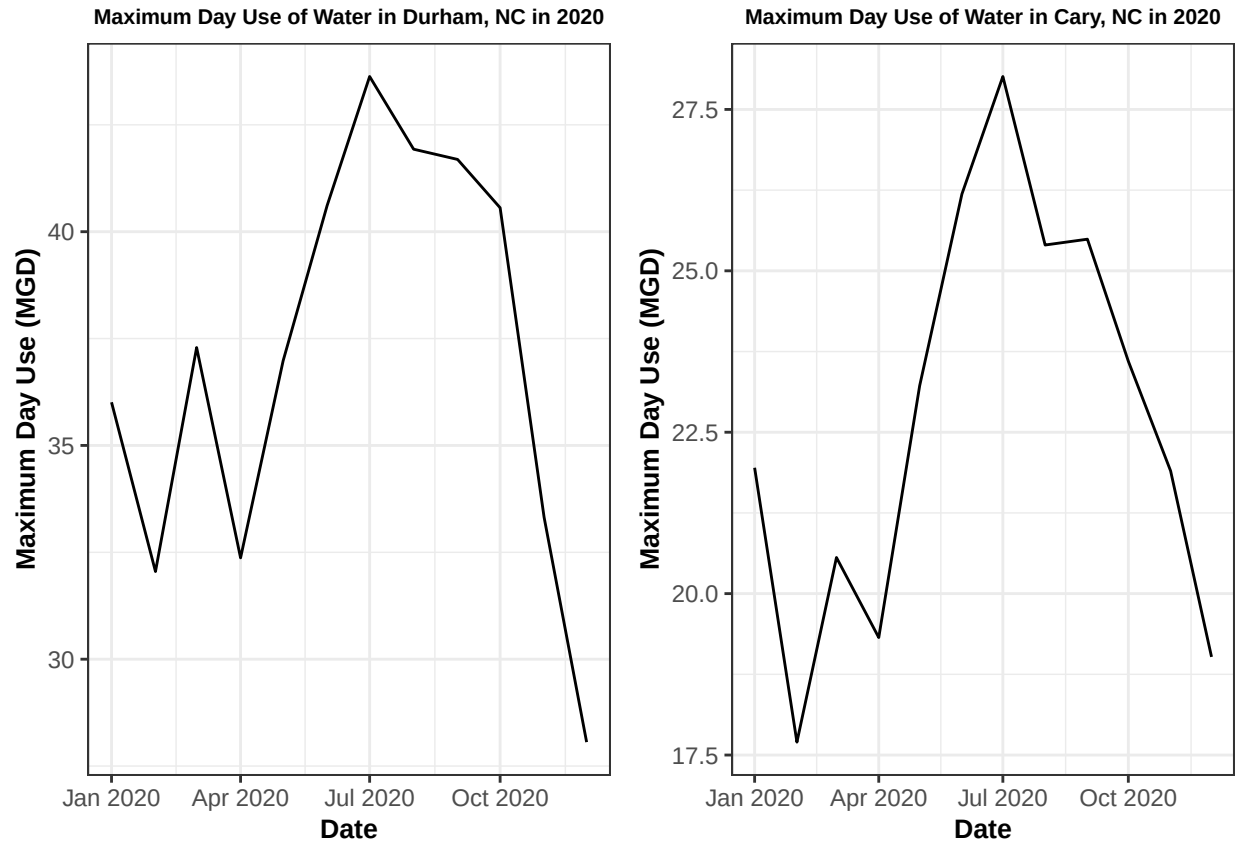
```
#8 using my scrape.it function to extract the max daily use for Cary for each
#month in 2020 and return it in a dataframe
Cary_2020_LWSP <- scrape.it.LWSP('03-32-010', 2020)

#plotting the date and maximum day use of water from the dataframe
Cary_2020_Plot <- Cary_2020_LWSP %>%
ggplot(aes(x= Date, y= maximum_daily_use_monthly)) +
  geom_line() +
  labs(title = "Maximum Day Use of Water in Cary, NC in 2020",
       x= "Date",
       y="Maximum Day Use (MGD)") +
  my.theme +
  theme(plot.title = element_text(face = "bold", size = 8, hjust = 0.5))

#viewing the plot
Cary_2020_Plot
```



```
#using cowplot to combine my Durham 2020 and Cary 2020 maximum day use of water plots  
plot_grid(Durham_2020_Plot, Cary_2020_Plot)
```



9. Use the code & function you created above to plot Cary’s max daily withdrawal by months for the years 2018 thru 2023. Add a smoothed line to the plot (method = ‘loess’).

TIP: See Section 3.2 in the “10_Data_Scraping.Rmd” where we apply “map2()” to iteratively run a function over two inputs. Pipe the output of the map2() function to bind_rows() to combine the dataframes into a single one, and use that to construct your plot.

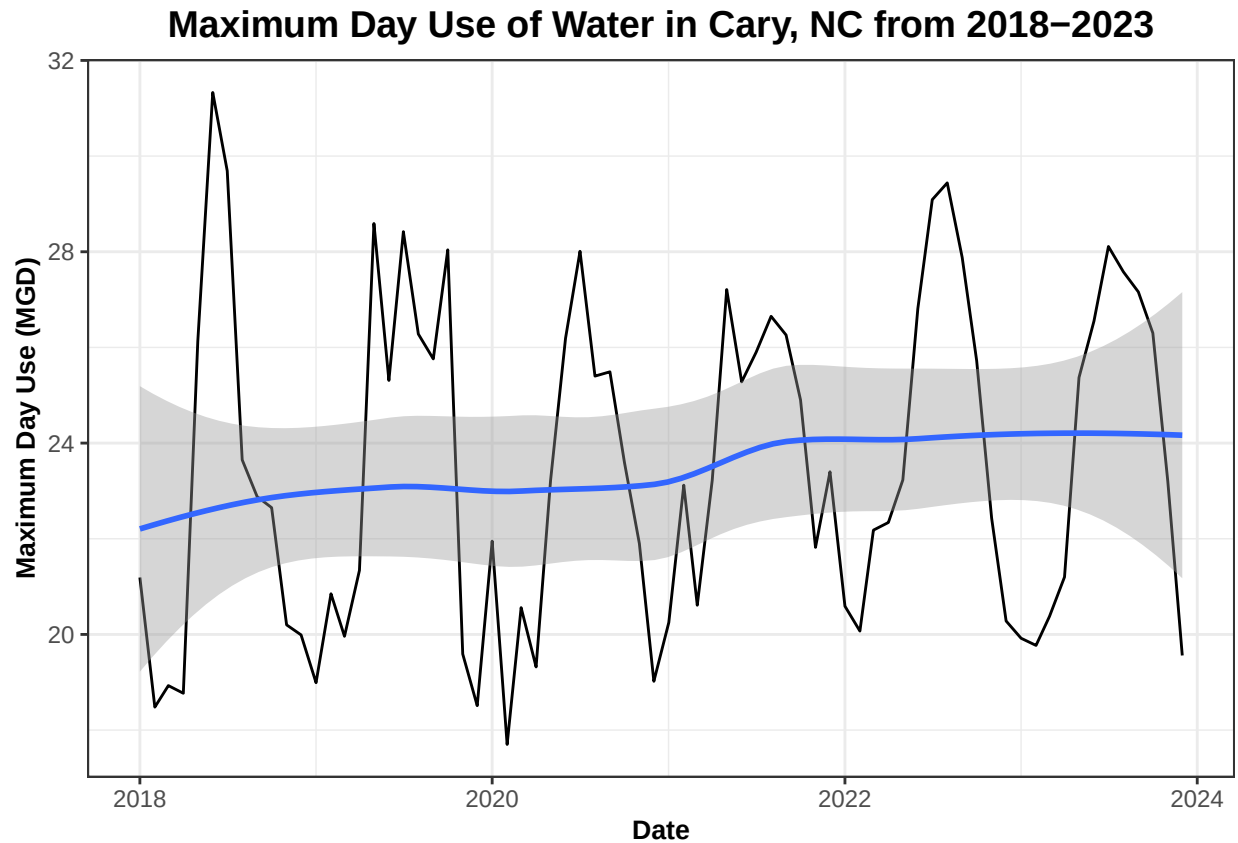
```
#9 creating a dataframe for Cary's maximum daily use by date for 2018-2023
Cary_2018_2023_LWSP <- map2('03-92-020', 2018:2023, scrape.it.LWSP)

#combining these into one dataframe
Cary_2018_2023_LWSP_Combo <- bind_rows(Cary_2018_2023_LWSP)

#creating a plot from the combined dataframe with a smoothed line
Cary_2018_2023_Plot <- Cary_2018_2023_LWSP_Combo %>%
  ggplot(aes(x= Date, y= maximum_daily_use_monthly)) +
  geom_line() +
  geom_smooth(method="loess") +
  labs(title = "Maximum Day Use of Water in Cary, NC from 2018-2023",
       x= "Date",
       y="Maximum Day Use (MGD)") +
  my.theme

#viewing the plot
Cary_2018_2023_Plot
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



Question: Just by looking at the plot (i.e. not running statistics), does Cary have a trend in water usage over time? > Answer: Yes, it does seem like Cary has a trend in water usage. The general trend is that usage is lower in the winter months, increases through the spring, peaks in the summer, and begins decreasing in the fall to reach the low winter point once again. >